G. Manohar

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Profile

Strategic Data Science leader with over 12 years of experience in AI/ML, fortified by 11 years of deep manufacturing expertise. I have successfully led the full lifecycle of AI initiatives, from concept to implementation, and consistently delivered complex solutions that solve critical business challenges across the automotive, oil & gas, and aerospace sectors.

Skills

Python	Linux	GIT	PyTorch	Keras
PowerBi	MLOps	Scikit-learn	Numpy	Pandas
OpenCv	Statistics	NLP	Azure	Six-sigma
LLM	AgenticAi	Automation	MLflow	Autoware
ROS2	Ubuntu	CAD	Paper replication	

Education

Advanced Certification in Data science & Artificial intelligence - IIT-Madras -Intellipaat 2022 - 2023
Bachelor of Mechanical Engineering - Satyabhama University 2010- 2014
Diploma in Mechanical Engineering - AVC Polytechnic College 2000- 2003

Trainings & Achievements

Autonomous Vehicle Systems Training by NITTTR Chennai Feb 2026 (Five Days)

AI Agents Intensive Course by Google

Gen AI Intensive developer by Google

Fundamentals of LLMs of The LLM Course By HuggingFace

Fine-Tuning of LLMs Course By HuggingFace

Best Performer : Sep-2012, Aug-2013, Feb-2015, Oct-2016 (Caterpillar Inc.)

Supplier Award of the Year: 2015 (Caterpillar Inc.)

Best Performer of the Year: 2016-17 (Quest Global)

Saved $2M - 2017-2018 (Caterpillar Inc.) by value Additions core change

On-the-Fly Award: Twice (Quest Global)

Certified Recruiter Training (Quest Global)

Experience's

1.PROGRAM MANAGER-AI – MSM - MARCH 2025 - PRESENT

- *AI Strategy & Innovation:* Defined and delivered global GenAI and Agentic AI roadmaps, translating RAG and multi-agent architectures into measurable ROI and board-level impact.
- *Program Leadership & Governance:* Led global AI/ML programs with 95%+ on-time delivery, implementing strong governance, risk controls, and value realization frameworks.

- *Platform Architecture & Technical Oversight:* Directed cloud-native AI deployments on Azure, establishing MLOps, LLM evaluation standards, and secure enterprise integrations.
- *Engineering & Organizational Development:* Scaled high-performing global teams, fostered ownership culture, reduced dependencies, and mentored senior leaders.
- *Data & Cloud Enablement:* Modernized enterprise data ecosystems using scalable pipelines and microservices to build compliant, production-ready AI platforms.
- *Growth & Strategic Partnerships:* Accelerated revenue through AI pre-sales, RFP leadership, and strategic global partnerships.
- *Operational Excellence & Compliance:* Established AI KPIs and Responsible AI frameworks ensuring quality, compliance, and lifecycle governance.
- *Stakeholder & Executive Engagement:* Partnered with C-suite and boards to align AI strategy with business readiness and enterprise-wide adoption.

Project	
E-commerce—AgenticAi (Lang chain framework)	Customer Req agent
	Stock clear Agent
	Transit management agent

2.SENIOR LEAD ENGINEER -QUEST GLOBAL -OCT 2013- JAN 2025

- Led end-to-end AI initiatives across the automotive, oil & gas, and aerospace sectors, driving strategic growth by delivering measurable business outcomes.
- Oversaw client relationships, business development, and P&L for complex AI and machine learning programs, ensuring project profitability and stakeholder satisfaction.
- Directed the full AI solution lifecycle, from conducting initial requirements analysis and developing proof-of-concepts to ensuring seamless deployment and operational success.
- Developed, tuned, and deployed complex AI/ML models, specializing in predictive analytics, forecasting, and computer vision for industrial applications.
- Architected and implemented scalable edge-to-cloud solutions, utilizing MLOps and LLMOps pipelines to ensure continuous delivery and robust performance.
- Led process optimization and efficiency enhancements using Agile and Six Sigma (DMAIC) methodologies, significantly improving supply chain and product development workflows.
- Monitored automation and model performance through metrics and reports, ensuring that quality, cost, and delivery objectives were consistently met.
- Directed and mentored a team of 10 ML engineers, overseeing project allocation, managing performance, and coordinating cross-functional, onshore/offshore development.

Algorithm worked & learned:

Task : Regression,Classification,Ensemble,Dimensionality Reduction,Clustering,Time Series, NLP

object detection & Classification, Multi-model, product based

Agentic/GenAI and Ecosystem:

Architecture : RAG based Systems,Model Fine-Tuning, Computer Vision

Models: OpenAi , Google Ai, HuggingFace Transformer

Supports: LangChain,LangGraph,LangSmith,FastAPI, Docker, Gradio, ChromaDB, FAISS.

Projects

PPRS -Parts Proliferation Reduction System	GenAI-Powered System for Cost Reduction Through Intelligent Prevention of Unnecessary New Part Creation.
EPI -Engine Parts Identifier (Computer Vision)	Engine component identification application powered by a fine-tuned EfficientNet model
GCE -Geometric Cost Estimation (Computer Vision)	Developed a YOLO-based object detection application for automated process cost estimation.
OSO -Optimized Service Offering	ML–driven optimization of supply chain costs, delivery timelines, and inventory storage efficiency.
CPPD- Continuous Product and Process Development	Architected a multi-module application integrating ML–driven capabilities across core functionalities.

3.LEAD ENGINEER -QUEST GLOBAL -NOV 2011– SEP 2013

- Monitored design changes and new releases per customer deadlines, allocating engineers based on project requirements.
- Validated designs for Caterpillar, Perkins, and Mak engine components, including fuel injection, cooling, lubrication, turbo with intercooler, and fluid line systems.
- Initiated Six Sigma projects and implemented automation solutions to improve processes, product quality, and reduce review cycles.
- Provided technical training and mentorship on design standards, fluid line systems, and engineering best practices.
- Led manufacturing automation, feasibility studies, and collaborated with coordinators, suppliers, and design managers on CPI and NPI projects, delivering measurable cost and process improvements.

4.PRODUCT DESIGN ENGINEER-ARMSTRONG INTERNATIONAL -JUN 2010 – OCT 2011

- Designed, modeled, and prepared detailed engineering drawings for pump traps, valves, and hot water modules compliant with PED regulations for European clients.
- Developed piston control valves for steam applications, localized steam products, and supported vendor development for casted and machined components.
- Coordinated production, process, and supplier deviations while preparing technical presentations, product catalogues, and animations for customer engagement.
- Conducted hydro, thermal, and structural testing as per IBR/ASME standards and prepared documentation to secure product approvals from IBR.

5.DESIGN ENGINEER (PROJECTS R &D) -SHIMI DESIGN (CLIENT L&T) -JAN 2006 – MAY 2010

- Designed and developed special products for domestic and international clients, covering sub-systems such as welded structures, shafts, springs, gears, belt drives, and bearings.

- Created 3D models and assemblies in SolidWorks/AutoCAD, validated designs using FEA/CFD, and ensured compliance with global standards (IS, DIN, ASTM, BS).
- Prepared fabrication and machining drawings, hydraulic/pneumatic schematics, and implemented routing modules for system integration.
- Coordinated with customers and production teams, presenting design results for approval and supporting precise articulation equipment development.

6.CAD ENGINEER -E-CORE ENGINEERS-JUL 2004 – DEC 2005

- Converted 3D models into 2D machining drawings, including reverse-engineered fixture components for Sundaram Fasteners.
- Designed jigs, fixtures, and associated systems for seal assembly in Nyloc nut manufacturing, including exploded view drawings for faster shop assembly.
- Coordinated with vendors for cost-effective fabrication, implemented fixtures on production lines, and conducted pilot runs to ensure quality consistency.

7.CNC PROGRAMMER/ SUPERVISOR -SWASTIKA INDUSTRIES-JUN 2003 – MAY 2004

- Programmed CNC machines, selected tools, and designed jigs/fixtures for high-precision pneumatic and hydraulic components, including complex castings.
- Implemented process improvements (e.g., automatic bar-feeding mechanism) to reduce machining time and improve delivery performance.
- Collaborated with cross-functional teams (ISO 9001 audits, material handling, quality) and coordinated with customers to meet technical and delivery requirements.